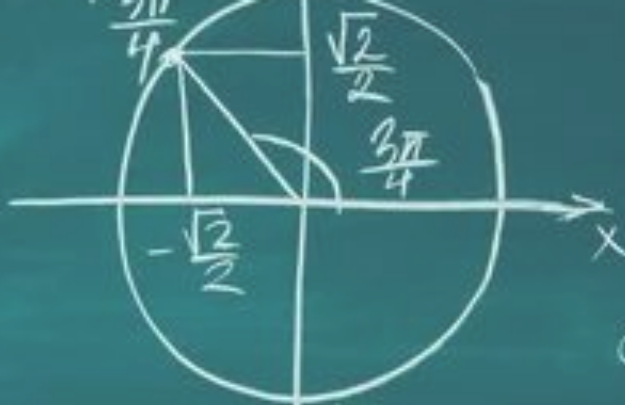
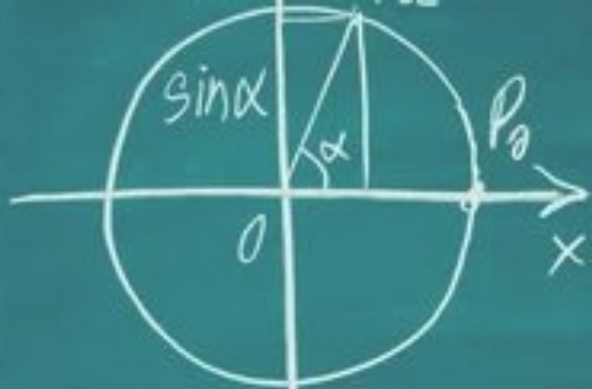




$$\operatorname{tg} \frac{3\pi}{4} = -1$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \operatorname{tg} \frac{3\pi}{4} = -1$$

$$D(\sin) = D(\cos) = 0 \quad E(\sin) = E(\cos) = [1; 1]$$

First year Aerospace Engineering

Integral Calculus Exercises

Marcos Carrera Valle

Find the integral of the following functions

- 1) $f(x) = 2.4x^3 - 3e^{(0.5x)}$
- 2) $f(x) = 5x^{(2/3)} + 4/(2x+1)$
- 3) $f(x) = 0.6\sin(3x) - 2x^{(-3)}$
- 4) $f(x) = 3\sec^2(0.2x) + 4e^{(-2x)}$
- 5) $f(x) = 7/\sqrt{x} - 2\cos(4x) + x^{(1.5)}$
- 6) $f(x) = 4\csc^2(2x) - 3x^{(-0.5)}$
- 7) $f(x) = 0.8e^{(4x+1)} + 5\sec(3x)\tan(3x)$
- 8) $f(x) = 6x^{(-1/2)} - 0.5\sin(0.1x)$
- 9) $f(x) = 2/(x^2) + 3\cos(0.5x) - 4e^{(x/3)}$
- 10) $f(x) = 1.5x^{0.5} + 4/(3x) - \csc(2x)\cot(2x)$

Find the integral of the following functions

- 11) $x * \cos(3x)$
- 12) $(2x + 1) / (x^2 + x + 0.5)$
- 13) $4 / ((x - 1)(x + 3))$
- 14) $x^2 * (x^3 + 5)^4$
- 15) $\cos^2(0.5x)$
- 16) $2x * e^{(4x)}$
- 17) $\sin(x) * e^{(\cos(x))}$
- 18) $e^{(2x)} / (1 + e^{(2x)})$
- 19) $\sin^2(x)$
- 20) $(3x + 1) / (x^2 - 1)$

Find the integral of the following functions

- 21) $3x * \ln(x)$
- 22) $x / \sqrt{2x^2 + 5}$
- 23) $\sec^2(x) / (1 + \tan(x))$
- 24) $10 * \sin(x) * \cos(x)$
- 25) $1 / (x^2 - 9)$
- 26) $x^2 * e^x$
- 27) $5x * \sqrt{1 - x^2}$
- 28) $1 / (x * \ln(x))$
- 29) $(\cos(x) + \sin(x))^2$
- 30) $(x + 5) / (x^2 + 5x + 6)$

Find the integral of the following functions

- 31) $4x * \sin(0.5x)$
- 32) $e^x * \sqrt{e^x + 1}$
- 33) $(\cos(x) - \sin(x)) / (\sin(x) + \cos(x))$
- 34) $\cos(2x) / (\cos(x) + \sin(x))$
- 35) $10 / (2x^2 + 5x - 3)$
- 36) $\ln(2x)$
- 37) $x^3 * \cos(x^4)$
- 38) $4x^3 / (x^4 + 0.2)$
- 39) $\sin^2(4x)$
- 40) $2x / ((x - 2)(x - 3))$

You are calculating the total volume of a wing for a new drone.

The wing has a span (b) of 10 meters, extending from $y = 0$ (root) to $y = 10$ (tip).

The Cross-Sectional Area (A) of the airfoil depends on the chord length (c) according to: $A(c) = 0.12 * c^2$

The Chord Length (c) tapers along the span (y) according to: $c(y) = 3 * e^{(-0.1y)}$

41) Calculate the total volume of the wing by integrating the Area function from $y = 0$ to $y = 10$.

A fighter jet is performing an interception maneuver.

The engines consume fuel at a rate (R) in kg/s, which depends on the engine Power (P) in kiloNewtons (kN): $R(P) = 0.05 * P$

The pilot increases Power linearly with time (t) in seconds: $P(t) = 200 + 4t$

42) The aircraft starts with 800 kg of fuel. If the maneuver starts at $t = 0$, find the exact Time (T) when the fuel tank becomes empty.

A rocket booster fires for 20 seconds.

The Thrust (F) produced by the engine is not constant; it decays as the combustion chamber pressure drops.

The Thrust in Newtons is given by: $F(t) = 500000 / (t + 10)$

43) Calculate the Total Impulse (J) delivered by the rocket stage.

(Total Impulse is the definite integral of Thrust with respect to time)

A C-17 Globemaster is winching a damaged tank up the rear loading ramp.

The tension in the cable (Force F) changes as the tank moves up the inclined ramp due to the changing angle of the cable.

The Force in Newtons is modeled by: $F(x) = 40000 + 2000 * x^{(0.5)}$

where x is the distance pulled in meters.

44) Calculate the total Work done (in Joules) to pull the tank a distance of $x = 9$ meters.

The "Perseverance II" rover relies on solar panels.

The Power (P) generated varies throughout the Martian day (sol) as the sun rises and sets.

The Power generation in Watts is modeled by a parabola: $P(t) = -2(t - 6)(t - 18)$

valid for the daylight hours $t = 6$ to $t = 18$.

45) Calculate the Total Energy (in Watt-hours) stored in the batteries during one day.

An engineer models the nose cone of a supersonic missile by rotating a curve around the x-axis. The profile of the nose cone is given by the function: $y = 0.5 * x^{(0.75)}$ for x between 0 and 4 meters.

46) Find the Volume of the nose cone generated by rotating this curve 360 degrees (2π radians) around the x-axis.

The outer casing (cowling) of a jet engine needs to be painted with a special heat-resistant coating. The shape is approximated by rotating the function: $y = 3 + 0.1x^2$ around the x-axis for the interval $x = 0$ to $x = 5$ meters.

47) Calculate the Surface Area of the cowling.

A de Laval nozzle (used in rockets) has a converging-diverging shape. The internal radius is modeled by rotating the function: $y = 0.2x^2 - 0.8x + 1.5$ around the x-axis, for the interval $x = 0$ to $x = 5$ meters.

48) Find the internal Volume of the nozzle.

A new type of omni-directional antenna is being designed. Its shape is formed by rotating the curve: $y = 2 * x^{(0.5)}$ around the Y-axis.

The curve is defined for x between 0 and 2 meters.

49) Find the Surface Area of this antenna

You are designing the cabin section of a business jet. In the blueprints, the floor of the hangar is defined as $y = 0$. The center axis of the fuselage is suspended at a height of $y = 3$ meters. The top outer profile of the fuselage is given by the function: $y = 3 + 2 * e^{(-0.3x)}$ for the section $x = 0$ to $x = 6$ meters.

50) Calculate the Volume of the fuselage.

SOLUTIONS

Find the integral of the following functions

- 1) $F(x) = 0.6x^4 - 6e^{(0.5x)} + C$
- 2) $F(x) = 3x^{(5/3)} + 2\ln(2x+1) + C$
- 3) $F(x) = -0.2\cos(3x) + x^{(-2)} + C$
- 4) $F(x) = 15\tan(0.2x) - 2e^{(-2x)} + C$
- 5) $F(x) = 14x^{(1/2)} - 0.5\sin(4x) + 0.4x^{(2.5)} + C$
- 6) $F(x) = -2\cot(2x) - 6x^{(0.5)} + C$
- 7) $F(x) = 0.2e^{(4x+1)} + (5/3)\sec(3x) + C$
- 8) $F(x) = 12x^{(1/2)} + 5\cos(0.1x) + C$
- 9) $F(x) = -2/x + 6\sin(0.5x) - 12e^{(x/3)} + C$
- 10) $F(x) = x^{1.5} + (4/3)\ln(x) + 0.5\csc(2x) + C$

Find the integral of the following functions

- 11) $(1/3)x \sin(3x) + (1/9)\cos(3x) + C$
- 12) $\ln(x^2 + x + 0.5) + C$
- 13) $\ln(x - 1) - \ln(x + 3) + C$
- 14) $(1/15)(x^3 + 5)^5 + C$
- 15) $0.5x + 0.5\sin(x) + C$
- 16) $0.5x e^{(4x)} - 0.125e^{(4x)} + C$
- 17) $-e^{(\cos(x))} + C$
- 18) $0.5\ln(1 + e^{(2x)}) + C$
- 19) $0.5x - 0.25\sin(2x) + C$
- 20) $2\ln(x - 1) + \ln(x + 1) + C$

Find the integral of the following functions

21) $1.5x^2 * \ln(x) - 0.75x^2 + C$

22) $0.5 * \sqrt{2x^2 + 5} + C$

23) $\ln(1 + \tan(x)) + C$

24) $-2.5\cos(2x) + C$

25) $(1/6)\ln(x - 3) - (1/6)\ln(x + 3) + C$

26) $e^x(x^2 - 2x + 2) + C$

27) $-(5/3)(1 - x^2)^{1.5} + C$

28) $\ln(\ln(x)) + C$

29) $x - 0.5\cos(2x) + C$

30) $3\ln(x + 2) - 2\ln(x + 3) + C$

Find the integral of the following functions

31) $-8x \cdot \cos(0.5x) + 16\sin(0.5x) + C$

32) $(2/3)(e^x + 1)^{1.5} + C$

33) $\ln(\sin(x) + \cos(x)) + C$

34) $\sin(x) + \cos(x) + C$

35) $(10/7)\ln(2x - 1) - (10/7)\ln(x + 3) + C$

36) $x \cdot \ln(2x) - x + C$

37) $0.25\sin(x^4) + C$

38) $\ln(x^4 + 0.2) + C$

39) $0.5x - (1/16)\sin(8x) + C$

40) $6\ln(x - 3) - 4\ln(x - 2) + C$

41) 4.67 cubic meters

42) 52.47 seconds

43) 549,306 Ns

44) 396,000 Joules

45) 576 Watt-hours

46) 100.53 cubic meters

47) 117.3 square meters

48) 27.0 cubic meters

49) 15.68 square meters

50) 20.37 cubic meters